Customer-service queuing based on predicted loyalty outcomes

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Abstract

We introduce a customer service management approach with a new type of predictive priority queuing. The priority rank of a case is determined based on predicted customer loyalty scores. Increases in customer loyalty are predicted based on predicted case throughput time. Case throughput time is, in turn, predicted from seasonality indicators alone, due to incomplete information at the time of entering the queue. The priority queue is continuously updated. Two studies are reported. Study 1 is a statistical analysis of two years of customer service data from a European internet and telecommunications provider. We utilise the real-world data to obtain valid estimates of the parameters of all distributions governing the service process. In Study 2, we use the calibrated distribution parameters in Monte Carlo simulations of the customer service management system. In these experiments, we compare our predictive priority queuing approach with the traditional first come, first served approach used in the case company. Furthermore, we compare the performance to two other approaches that are based on predicted case throughout time; the longest remaining time first and the shortest remaining time first approach. We show that prioritisation based on predicted customer loyalty does indeed lead to an increase in average customer loyalty. However, since this effect is mediated by predicted throughput time, the longest remaining time first approach yields similar results as our proposed method. Introduction of a service level (here: maximum 60 hours waiting time) reduced the performance of all approaches based on predicted throughout time to the performance of the first come, first served approach.

Keywords: Prescriptive analytics, Customer service, Dynamic queue prioritisation, Net promoter score

1. Introduction

In a classical experiment, [1] show that service time guarantees can be a two-edged sword: customers will only be more satisfied at the end of a waiting period if the guaranteed service times are actually met. If the guarantees are not met, however, customers will be even less satisfied than if they had not been aware of the existence of a service time guarantee in the first place. This poses a challenge for customer service management [2]. Since service processes are interactive by nature, depending to a large degree on actions by the customer, they can never be fully under the control of the service provider. Compared to manufacturing processes, a much larger proportion of cases will be outside the ”specification limits” defined by waiting time guarantees [3]. The problem is known and has been addressed in several ways in previous research.

Service management approaches based on queuing theory, for example [4], [5] or [6], reduce the between-case variance of the waiting time distribution and thereby also the number of cases exceeding a given waiting time guarantee. In its simplest form, a queuing system treats newly arriving cases as equal in terms of the gains associated with meeting a waiting time guarantee or the losses associated with exceeding a waiting time guarantee. This is for example the case with the standard first come, first served (FCFS) discipline. More elaborate approaches introduce classes of cases or customers in order to prioritise the queue [7]. Class membership can be static and known before case arrival (e.g., based on customer tenure, customer lifetime value or a related metric). Alternatively, the class membership of a case may be dynamic but known at the time the queue is updated (e.g., based on current prices of spare parts or repair activities). Finally, it can be dynamic and unknown at the time the queue is updated. In this situation, a prediction at case level is required of the quantity (or quantities) based on which priority class membership is to be assigned.

In the research presented here, we shall introduce such an approach. We study an $M/M/s$ system with priority queuing. The priority rank of a case is determined based on predicted increases in customer loyalty. Increases in customer loyalty are predicted based on predicted case throughput time. Case throughput time is, in turn, predicted from seasonality indicators alone, as no case attributes are available at the time of prioritisation. The priority queue is updated every time a new case arrives. Using a simulation approach calibrated on two years of real-world customer service data, we show that prioritisation based on predicted increases in customer loyalty does indeed lead to an increase in average customer loyalty (compared to first come, first served). In addition, it also leads to lower average throughput time. Both effects are relatively stronger when the number of customer service agents s is lower. We also compare our approach to two other queuing disciplines two other approaches that are based on predicted case throughout time, including the longest remaining time first approach (LRTF) and the shortest remaining time first approach (SRTF). Compared to these, our loyalty-based approach still yields higher average customer loyalty, but only is only slightly better than SRTF, and equal to LRTF, in terms of case throughput time. Introduction of a service level (here: maximum 60 hours waiting time) to increase the fairness of the systems reduced the performance of all approaches based on predicted throughout time to the performance of the first come, first served approach used in the case company.

To our knowledge, this is the first paper in the queuing literature that addresses *predicted loyalty outcomes* in its prioritisation approach. The paper is structured as follows. In the remainder of the introduction section, we will review relevant theory and previous research, describe the rationale of our approach, and explain our research question and the case background for the empirical part. Then, we will report Study 1, a statistical analysis of two years of real-world customer service data from a European internet and telecommunications services provider. We will document the estimation of all distribution parameters needed in the simulation and the development of the statistical prediction models. In Study 2, we will use the calibrated distribution parameters and statistical models in simulations of our $M/M/s$ system, comparing our prediction-based prioritisation approach with the traditional FCFS approach and two other approaches based on predicted case throughout time: The SRTF and the LRTF approach.

1.1. Theory and previous research

1.1.1. Prioritisation approaches in queuing

In the terminology of [8], the "discipline" of a queuing system denotes the order in which the cases in the queue are processed. Traditional approaches include first come, first served (FCFS),

last in, first out (LIFO) and shortest job first (SJF). *Priority queuing* is used when the arriving cases differ in terms of their value to the business. Many prioritisation approaches have been suggested in the literature. In principle, prioritisation can be based on any set of cost or value indicators that are available on the level of a case or customer. [9], for example, study the impact of different prioritisation approaches in service systems. They propose a model based on an $M/M/s/N$ queuing system with the goal to find the optimal staffing s, given customer classes with different processing times, subject to a desired service level. The queue is prioritised using an idle-server based threshold priority and their proposed single-class staffing rule. The authors find that there is a decoupling between staffing and prioritisation approach when they evaluate a multi-class system, such that staffing in this setting has similar dynamics to a single class system.

In another paper, [10] study the impact of prioritised queue management under the assumption that prioritisation comes with a cost. Their approach features two types of agents: sorters and processors. The role of the sorters is to gather information and prioritise each of the incoming customers, while the processors perform the work needed on the cases. This introduces an additional cost related to the sorter agents, but assuming heterogeneous waiting costs, sorting is expected to have a net positive effect on total costs. The results show that, given a fixed exogenous budget, this prioritisation will only outperform FCFS in some cases, depending on the ratio of sorting cost to processing cost. [11] propose a queue prioritisation method based on a fuzzy system. In theory, the approach would assign weights to customers that are based on service duration, service value, customer tenure, customer patience, and waiting time. Based on (not empirically calibrated) simulation experiments, the authors find that their approach outperforms FCFS in an $M/M/1$ system with different levels of stability.

The work of [12] utilises the shortest remaining time first (SRTF) discipline in a case study of a hospital emergency department. To improve the patient flow, three variants of the SRTF discipline are compared to FCFS in calibrated simulation experiments of a $M/M/s$ queue. The authors find that the SRTF-based strategies lead to shorter throughput times. However, this depended heavily on the accuracy of the remaining time prediction. Furthermore, the authors report that generic SRTF had the disadvantage of leading to starvation, i.e. the unfair situation that some customers/patients will wait indefinitely in the queue). In [13] the authors study the application of multiple variants of the longest remaining time first discipline (LRTF) in the context of edge computing server allocation. Four different variants of the LRTF discipline are compared to the FCFS in a $M/M/s$ queue. The LRTF discipline led to the shortest average response times, compared to FCFS. However,in a similar manner as the SRTF discipline, generic LRTF suffers from the problem of starvation.

In the research presented in this paper, we will compare our proposed approach (NPS) to three competitors: the queue discipline most commonly applied in customer service systems (FCFS) plus two disciplines which are, like our proposed approach, based on predicted throughput times (SRTF and LRTF) but do not use the throughout time predictions in further prediction steps. We will compare the four disciplines in scenarios with and without waiting time guarantees in the form of a *service level*. Service levels are often introduced in customer service systems to avoid the starvation problems which disciplines such as SRTF and LRTF are known for (see above) and thereby to increase the fairness of priority-based queuing systems.

We will not consider preemptive queuing approaches in our comparisons. The concept refers to a group of queue disciplines (e.g., see [14] where a customer who arrives while the server is busy serving another customer is given priority over the existing customer. The server interrupts the ongoing service to attend to the new customer, and then resumes the previous service where they left off. This approach has some theoretical advantages, but it is rarely used in customer

service queuing systems. There are two reasons for this. First, customer service queuing systems are designed to provide fair and equitable service to all customers. Preemptive queuing strongly favors high-priority customers at the expense of low-priority customers, and often in quite obvious ways, which can lead to dissatisfaction and complaints. Second, customer service queuing systems are usually designed to optimise resource utilisation and minimise costs. Due to the frequent task-switching, preemptive queuing can increase the complexity and cost of the system, which may not be justifiable for the marginal gains in efficiency or responsiveness.

1.1.2. Customer loyalty as a prioritisation criterion

To our knowledge, customer loyalty has so far not been used in any prioritisation approaches studied in the queuing literature. This is somewhat surprising, considering that customer loyalty is the central construct in all relationship-oriented approaches to marketing and service management [15, 16, 17]: the more loyal a customer, the higher that customer's lifetime value, and the more loyal the customers of a company, the higher the company's expected future profitability. The commonly accepted definition of customer loyalty is "the biased (non-random) behavioural response (brand support), expressed over time by some decision-making unit with respect to one or more alternative brands out of a set of such brands, and is a function of psychological processes (brand commitment)" [18].

Whilst there is little dispute about the conceptual definition and the importance of loyalty, its operationalisation differs vastly in practice. Systematic reviews distinguish two basic approaches to the measurement of customer loyalty [19, 18, 20, 21, 22]: behavioural approaches and attitudinal approaches. Behavioural approaches to the measurement of loyalty use objective indicators. Typical measures are the share of purchases of a brand relative to the total purchases within the respective category, measures of allegiance (i.e., how long the customer has stayed with the supplier), and the number of competing suppliers a customer has relations with. Attitudinal approaches to the measurement of loyalty, on the other hand, use measures of commitment and intention, for example intentions to remain a customer, re-purchase the product or service, or recommend the product or service to others.

Unfortunately, attitudinal and behavioural measures of loyalty have often shown limited convergent validity. In a cross-category study of brand loyalty in consumer goods markets, [23] found a correlation of 0.64 between the two alternative measures. In a similar study, [24] found a much lower correlation of 0.23. In a study of loyalty in telecommunications service markets, [25] found correlations ranging from 0.10 to 0.50. Since its first publication by Bain consultant Frederick Reichheld [26], the *net promoter score* (NPS) has come to replace virtually all other attitudinal measures of customer loyalty. The NPS is based on a single survey question that measures word-of-mouth intention ("How likely is it that you would recommend [company or brand] to a friend or colleague?") on an 11-point Likert scale ranging from "not at all likely" (0) to "extremely likely" (10). At the analysis stage, the raw scale score is transformed into a three-class variable: customers who responded with levels 9 or 10 are labelled *promoters*, customers who responded with levels 6 or lower are labelled *detractors*, and customers who responded with levels 7 or 8 are labelled *passives*. The transformation is non-linear and follows the logic of the top-box rule in market research [27], acknowledging that only differences in the upper regions of intention scales have predictive validity with respect to differences in the probability of the target behaviour.

Due to their as-yet undecided status, the passives are often considered the segment where targeted customer relationship management activities can have the highest leverage and impact. Converting passives to promoters would require no more than a shift of one or two scale points

upwards on the response scale of the NPS survey question. In a similar way, passives could become detractors by a shift of one or two additional scale points downwards on the response scale. Hence, the passives can be regarded as the most sensitive segment of customers. We adopt this way of thinking in the prioritisation approach investigated here.

Let i ($i = 1, 2, ...N$) be an active case in queue Θ at time t ($t = 1, 2, ...T$). Furthermore, let $N\hat{P}S_i$ be the predicted NPS at time $t = T$ of the customer who submitted i, that is, after the service has been completed and the case has been resolved. The priority rank of case i is then determined by rank-transforming the distance of the predicted after-service NPS of the customer to the midpoint of the interval on the net promoter score response scale that represents the segment of the passives. This midpoint is $NPS = 7.5$, such that:

$$priority(i, t) = rank(|N\hat{P}S_{i,T} - 7.5|) \tag{1}$$

The rank function maps the set of NT distinct values to the set of integers $1, 2, ..., N$ such that each value is assigned a unique rank based on its position in the ordered list of values (i.e., the smallest value receives a rank of 1, the second smallest receives a rank of 2, and so on). The variable *priority* is therefore ordinal: a value of 1 indicates that case i has first priority at time t, a value of 2 indicates that case i has second priority at time t, and so on.

Since customer satisfaction and loyalty are partly determined by the customer's experience with the company's service provision, they will partly depend on the time it will take the company to resolve the case [28, 29, 30, 1, 31]. Since the throughput time of the case is not known before the case has been resolved, a prediction of the throughout time for case i must be made before the after-service customer loyalty $N\hat{P}S_{i,T}$ can be predicted.

1.2. Research questions

The objective of the research presented here is to develop a new approach to customer service management that addresses *customer loyalty* in its queue prioritisation approach, as well as evaluate its suitability via a case study. To achieve this, we divide the research into two studies:

In Study 1, we will document the estimation of all distribution parameters needed in a simulation model of the case company (which will be introduced in the next section). Furthermore, we will document the estimation of the needed prediction models in order to perform the proposed loyalty-based queue prioritisation. The work is guided by the following research question:

- **RQ1.1:** What are the distribution parameters of the key process components needed to generate an agent-based simulation model of the service process in the case company?

In Study 2, we will use the calibrated distribution parameters and statistical models in Monte Carlo simulations, comparing the NPS-based predictive priority queuing approach with three other queue disciplines (FCFS, LRTF, SRTF), guided by the following research questions:

- **RQ2.1:** How does the number of agents influence the queue waiting time in the simulation period under different prioritisation schemes?
- **RQ2.2:** What are the effects of the four queue disciplines on overall process performance?
- **RQ2.3:** What are the effects of the proposed loyalty-based queue discipline on the simulated net promoter score?

A conceptual overview of the two studies is shown in Figure 1. As illustrated in the figure, Study 1 utilises event log data provided by the case company which will be further introduced in section 1.3. Since neither case throughput time nor after-service customer loyalty are known at the point in time where cases must be prioritised, our approach involves two statistical prediction models: (a) a model that predicts expected throughput time, given case attributes that are known at the time the case enters the queue, and (b) a model that predicts customer loyalty, given throughput time and case attributes that are known at the time the case enters the queue. In each discrete time unit, both models are used to update the ordering of cases in the queue.

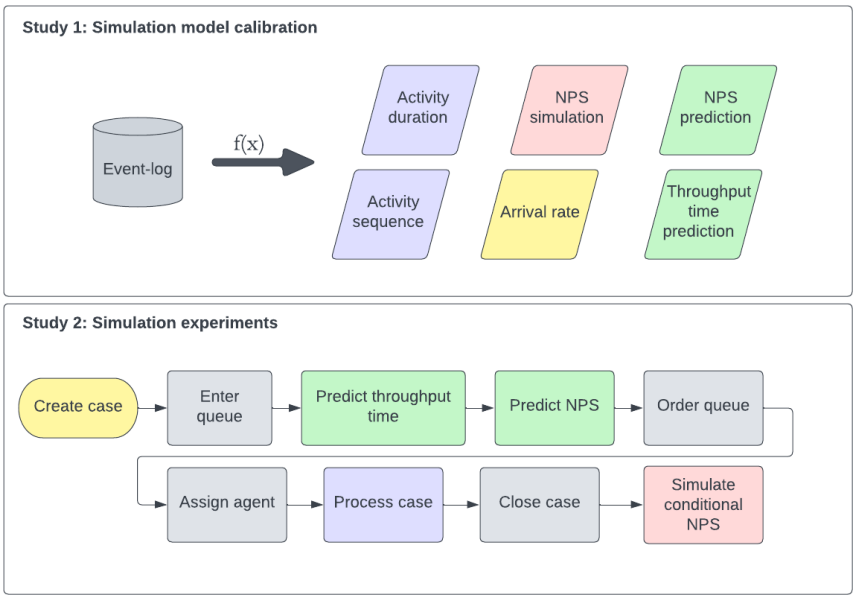

Figure 1: *Conceptual overview of the two studies.*

We see it as important that the merit of our approach is assessed based on as realistic assumptions as possible. Hence, the values of all simulation parameters will be based on a real business case. The case context as such is described in the following section.

1.3. Case context

The case company is a European internet and telecommunications services provider that wishes to remain anonymous. The case company uses a customer relationship management (CRM) system for the management of customer service requests from its enterprise customers. In the study period of almost two years (February 2018 to December 2019), the case company had 91 different agents working on 11,294 email-based service requests in total.

The agents are based at different physical locations, but have remote access to the same systems to help solve customer inquiries. Agents are organised into multiple teams, and do not exclusively work on customer service inquiries at all times. Customers can contact the company via e-mail, telephone or a customer support request form on the company's website. Contact via telephone is mainly utilised by medium and large enterprise VIP customers (these have their own

dedicated key account manager and thereby represent a minority of the customer base). Once a case is created, it is put in the queue until it is assigned to an available agent. Due to the form of contact, the case topic is either missing or unverified at the time the agent is assigned. Once an agent has been assigned, he or she will work on the case and have correspondence with the customer via e-mail or telephone, in case further information is needed to resolve the issue. When the issue is resolved, the case is closed.

The case company uses the Net promoter score (NPS) [26] as a tool to evaluate its performance after a case has been closed. The customer receives a short text message (SMS) and is asked to reply on a scale from 0 to 10, how likely he or she is to recommend the case company to others. Of all 11294 cases in the study period, 1897 received a valid NPS reply (16.7% response rate). The average individual NPS response was 8.82. The issue-to-resolution process at the case company is visualised as a BPMN model in Figure 2.

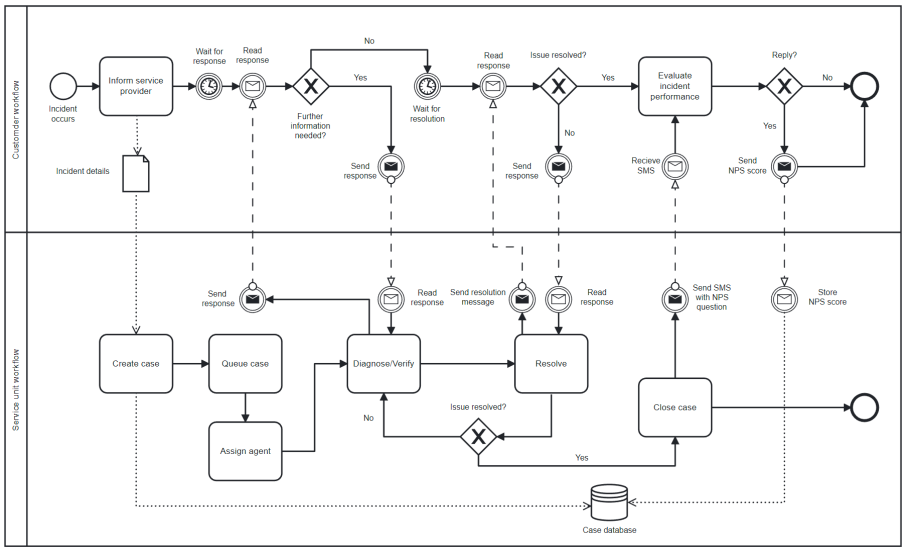

Figure 2: *BPMN model of the customer service process in the case company.*

2. Study 1: Statistical analysis of real-world data

Study 1 is a statistical analysis of the customer service process that will be simulated in Study 2. The purpose of the analysis is to obtain valid estimates of all distribution parameters governing the process: activity duration, activity sequences, inter-arrival times, case throughput time, and customer loyalty (operationalised in terms of the net promoter score). The main motivation for modelling activity duration, activity sequences and inter-arrival times is to enable good calibration to the real-world customer service process in the case company. The throughput time model is a key component in the predictive prioritisation approach used for queue management in Study 2. The net promoter score, which is dependent on throughput time, will be modelled in two variants: (a) including all relevant information, including information that will only be known once a case is completed, (b) using the predicted throughput time only. This will be further explained in Study 2.

2.1. Method

2.1.1. Data

The analysis will be performed on a subset of the data (1897 out of 11294 cases) from the module of the case company's Salesforce™ CRM system that is used for customer service management. All cases are created via email, and the subset contains only cases where, after the completion of the case, the relevant customer responded to the SMS request to answer the net promoter score (NPS) question. The data is in event log format (see 32) and consists of 5456 events related to 1897 cases. The event log was created by combining the Salesforce™ *case* and *task* objects. The case object contained static features such as the responsible agent, as well as the case topic and time stamps. The task object included the events related to the cases. At this level, an *interaction* with a customer denotes a phone call, whereas e-mail correspondence is explicitly coded as *email*.

2.2. Results

In the following, the key characteristics of the customer service process in the case company will be modelled using statistical techniques.

2.2.1. Activity duration

Modelling the conditional activity duration is a crucial input to the Monte Carlo simulations in Study 2 where the simulated throughput time will be modelled as a function of the waiting time plus the individual activity durations in a given case.

Activity duration was modelled using a set of model candidates from the generalised linear model family with different link functions and regularization penalties. An overview of the fitted models is shown in Table F.8. The selected model was a generalised linear model with a Weibull link function, which yielded a generalised $R^2 = 0.21$ on the training set (70-30 split) and a generalised $R^2 = 0.19$ on the test set. The prediction expression has the following form:

$$\hat{duration} = \exp\left(\alpha + \sum_{i=1}^{n} X_i\beta_i\right)\Gamma(1+\theta) \tag{2}$$

where X_i denotes the ith input feature, β_i the ith model parameter, Γ the *gamma-function*, and θ the shape parameter of the gamma distribution. The simulation equation is of the form:

$$duration = Weibull\left(\frac{1}{\theta}, \exp\left(\alpha + \sum_{i=1}^{n} X_i\beta_i\right)\right) \tag{3}$$

The model included features representing four input variables: *case topic*, *activity*, *activity number* and *resource*. Case topic refers to one of ten anonymised categories of customer service cases distinguished by the case company. Activity refers to one of three types of activities logged in the CRM system (see Section 2.1.1). Activity number is the order of the given activity in the trace. Resource is a categorical variable with 77 different levels referring to the individual agents observed in the two-year span of the data. This particular variable was forced into the model without regularisation in order to ensure that the effect of the individual agent (reflecting different speeds) would be represented in the model.

To enable further generalisation in the manner of a random effect, a normal distribution was fitted to the individual model coefficients representing the 77 different resource levels, with mean $\mu = 0.22$ and standard deviation $\sigma = 0.52$. The individual effects of the agents in Study 2 will

Table 1: Model coefficients: Weibull regression (target variable: activity duration in hours; square brackets represent levels of categorical factors; the last level is the reference level)

Term	$\hat{\beta}$	SE	p
Intercept	1.6645	0.1650	¡.0001
case_topic[d_2-z_4]	0.0200	0.0456	0.6620
case_topic[g_1-z_4]	-0.0538	0.0313	0.0857
case_topic[j_1-z_4]	-0.0557	0.0213	0.0088
case_topic[q_3-z_4]	0.1712	0.0587	0.0035
case_topic[r_2-z_4]	0.0836	0.0383	0.0288
case_topic[w_1-z_4]	-0.0609	0.0286	0.0334
case_topic[w_2-z_4]	0.0119	0.0313	0.7043
case_topic[z_2-z_4]	-0.0420	0.0433	0.3319
case_topic[z_3-z_4]	0.1637	0.0369	¡.0001
task_tasksubtype[Email-Task-Reminder]	0.0180	0.1158	0.8767
task_tasksubtype[Interaction-Task-Reminder]	0.1057	0.1319	0.4229
task_number	0.0420	0.0177	0.0176
resource	0.2171		
theta	0.3908		

Table 2: Transition matrix: Absorbing Markov chain.

	Task-Reminder	Interaction	Email	END
Task-Reminder	0.08	0.0	0.67	0.25
Interaction	0.0	0.02	0.96	0.02
Email	0.0	0.02	0.45	0.53
END	0.0	0.0	0.0	1.0

therefore be simulated from the fitted distribution, rather than the individual resources observed in the data.

The coefficients of the model are presented in Table 1. Since the *resource* effect is modelled using a normal distribution, its 77 original parameter values are excluded from the coefficients in Table 1. However, the full table with all parameters can be found in Appendix G.9.

2.2.2. *Activity sequences*

The *trace* or sequence of activities in a given case is modelled using an absorbing Markov chain. The model thus consists of $k + 1$ states, where the states are the possible activities plus the absorbing state, 'END'. Here, the absorbing state is added as the last step in each of the observed traces. The Markov chain prediction has the general form:

$$P(X_{t+1} \mid X_1, X_2, ..., X_t) = P(X_{t+1} \mid X_t) \tag{4}$$

The transition probabilities P were estimated using maximum likelihood and are shown in Table 2, whilst the initial probabilities P^0 are shown in Table 3.

2.2.3. *Inter-arrival times*

The inter-arrival times are the times between the arrivals of two consecutive cases. Modelling this distribution is important for Study 2 where the inter-arrival times control the traffic intensity

Table 3: Initial probabilities

	Task-Reminder	Interaction	Email	END
P^0	0.0	0.92	0.08	0.0

Table 4: Model coefficients: exponential regression (target variable: hours to next case arrival)

Term	$\hat{\beta}$	SE	p
Intercept	726.6267	323.9276	0.0249
year	-0.3589	0.1605	0.0253
month	-0.0881	0.0232	0.0001
day	0.0078	0.0081	0.3308
weekday	0.2616	0.0473	¡.0001

in the modelled system (here a customer service function), given different levels of staffing. As the simulation will run over a one-year period between 2018 and 2020, we include seasonality patterns in the model.

A set of generalised linear models with different link functions was estimated. The candidate models are shown in Table H.12 in the Appendix. The selected model was a generalised linear model with an exponential link function, which yielded a generalised $R^2 = 0.33$ in the training set (70% of the data) and a generalised $R^2 = 0.34$ in the validation set (30% of the data). The prediction expression has the following form:

$$\widehat{InterArrivalTime} = \exp\left(\alpha + \sum_{i=1}^{n} X_i\beta_i\right) \tag{5}$$

The simulation equation is:

$$InterArrivalTime = -\log(1 - U(0,1)) \times \exp\left(\alpha + \sum_{i=1}^{n} X_i\beta_i\right) \tag{6}$$

Where $U(0,1)$ represents a scalar from the uniform distribution with $min = 0$ and $max = 1$. The model coefficients are shown in Table 4

2.2.4. Throughput time

The main purpose of modelling case throughput time is to provide additional information about a given case when it has not yet been assigned. This information will be used to model the NPS, which is used to prioritise the cases. A set of candidate models was estimated (see Table I.13). The selected model was a generalised linear model with an exponential link function, which yielded a generalised $R^2 = 0.055$ in the training set (70% of the data) and a generalised $R^2 = 0.053$ in the validation set (the remaining 30% of the data). The target was modelled with an offset of 1. The prediction expression has the form:

$$\widehat{Throughput_time} = exp(\alpha + \sum_{i=1}^{n} X_i\beta_i) - 1 \tag{7}$$

Estimates of the model coefficients are shown in Table 5. The features represent the date and time of a case's arrival in the queue. We deliberately used simple linear approximations to the

Table 5: Model coefficients: exponential regression (target variable: case throughput time in minutes)

Term	$\hat{\beta}$	SE	p
Intercept	66.809	355.374	0.851
year	-0.030	0.176	0.867
month	-0.041	0.028	0.144
day	0.000	0.000	1.000
weekday	0.000	0.056	0.995
hour	0.065	0.052	0.211

nonlinear effects of hour of day, weekday, and day of month in order to make the model more generalisable beyond this particular customer service group in our case company. The linear features model more general effects such as different arrival rates and case resolution times at the beginning versus end of work day and work week [33, 34, 35]. Even though this approximate model may have low precision in its predictions of specific durations, these predictions will still be sufficient for assigning priority ranks to cases. In the decision making literature, this property is known as the "robust beauty of improper linear models" [36].

2.2.5. NPS

The NPS is modeled for two specific purposes: (a) Simulating the NPS given complete information about the case after it has been closed, and (b) generating an initial prediction of the expected NPS using incomplete information. This section will first present the conditional distribution which will be used for simulating the NPS given complete information. A set of candidate models were estimated from the family of generalised linear models with normal, gamma and exponential link functions. The results are shown in Table J.14 in the Appendix. The selected generalised linear model with gamma link yielded a generalised $R^2 = 0.040$ in the training set (70% of the data) and $R^2 = 0.039$ in the validation set (the remaining 30% of the data). The target variable was modelled using an offset of 1 (as the gamma distribution cannot represent zero). The simulation equation has the form:

$$NPS = \Gamma(\frac{Exp(\alpha + \sum_{i=1}^{n} X_i\beta_i)}{\rho}, \rho) - 1 \tag{8}$$

Where $\Gamma(k, \rho)$ denote the gamma distribution with shape parameter k and scale parameter $\rho = 1.3005$. The shape parameter is conditional on the intercept α and the linear combination of the input features X and their associated coefficients β, which are shown in Table 6.

For the initial NPS prediction model, which will be used to prioritise incoming cases under incomplete information, the case topic will not be available before the case has left the queue. We therefore only use predicted throughput time to predict the NPS value. In this case, the best model candidate was linear as shown in Equation (9).

$$N\hat{P}S = (\alpha + x\beta) - 1 \tag{9}$$

This model yielded R^2 values of 0.01 and 0.03 in the training and validations sets, respectively. The model coefficients are shown in Table 7.

Table 6: Model coefficients: gamma regression (target variable: NPS)

Term	$\hat{\beta}$	SE	p
Intercept	2.3006	0.0430	<.0001
Log[case_throughputtime+1]	-0.0098	0.0052	0.0590
case_topic[d_2-z_4]	-0.1291	0.1049	0.2183
case_topic[g_1-z_4]	0.1008	0.0481	0.0362
case_topic[j_1-z_4]	0.0853	0.0433	0.0490
case_topic[q_3-z_4]	-0.0427	0.0549	0.4359
case_topic[r_2-z_4]	-0.0317	0.0848	0.7088
case_topic[w_1-z_4]	0.0000	0.0000	1
case_topic[w_2-z_4]	0.0000	0.0000	1
case_topic[z_2-z_4]	0.0967	0.0509	0.0577
case_topic[z_3-z_4]	-0.0048	0.0484	0.9217

Table 7: Model coefficients: gamma regression (target variable: NPS)

Term	$\hat{\beta}$	SE	p
Intercept	10.2211	0.1111	¡.0001
Log[case_throughputtime+1]	-0.0949	0.0249	0.0001

3. Study 2: Monte Carlo experiments

In Study 2, we will recreate the customer service process in the case company in a simulation model, using the results of Study 1 to calibrate the relevant model parameters. In Monte Carlo experiments, we then compare the performance of the loyalty-based predictive priority queuing approach suggested in this paper with the traditional first come, first served approach. An overview of the approach is shown in Figure 3.

For the loyalty-based queue management approach, a *priority score* is calculated every 15 minutes (update interval of queue management and case assignment) as: $Priority_score = |7.5 - NPS_{arrival}|$. Here, $NPS_{arrival}$ refers to the predicted NPS-score (see 2.2.5) based on estimated throughput time (see Section 2.2.4) before the case has been assigned.

To evaluate the outcome of the experiments we also define NPS_{resp} which is the simulated conditional NPS-response, based on the case type and waiting time. To compare the two approaches, we run a set of simulation experiments spanning over 365 days in the period from 2018/07/01 to 2019/07/01.

3.1. Method

The method section is structured as follows: Firstly, the overall experimental design is presented in Section 3.1.1. Next, an overview of the simulation framework is presented. Sections 3.1.3 to 3.1.9 present details of the algorithms used in the experiments.

3.1.1. Experimental design

The simulation experiments were generated according to a full factorial $4 \times 7 \times 2$ design. The first factor was the queue discipline, the second the number of agents available, and the third the service policy. An overview of the three factors and their levels is shown below.

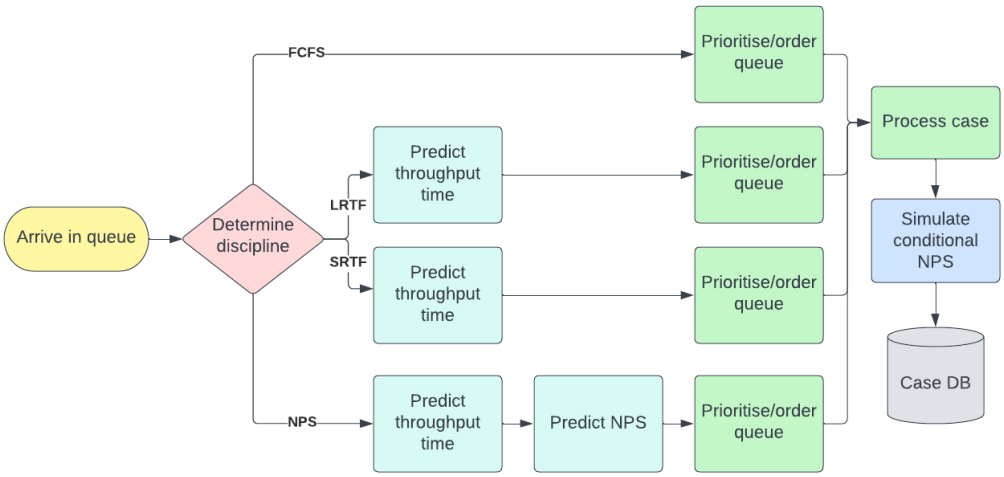

Figure 3: Overview of the competing queue management approaches.

- **Factor 1: Queue discipline**
  - FCFS - First come, first served
  - SRTF - Shortest remaining time first
  - LRTF - Longest remaining time first
  - NPS - Net promoter score-based priority
- **Factor 2: Number of agents**
  - 3, 4, 5, 6, 7, 8, 9
- **Factor 3: Service level constraint**
  - None
  - Service level = 60 hours

For the queue discipline we use non-preemptive queuing and benchmark our NPS-approach against $FCFS$, $SRTF$ and $LRTF$. $FCFS$ is the current approach used in the case company and thus serve as the baseline. As no other attributes for case prioritisation are available when a case arrives (the topic category is only realised when an agent has read the email), the $SRTF$ and $LRTF$ approaches are implemented based on predicted remaining throughput time (Eq. 7). The $SRTF$ and $LRTF$ disciplines are included to provide contrast to our approach, illustrating two alternative predictive scheduling approaches using a similar input. The range in the number of agents is chosen to represent the contrast between sufficient and insufficient capacity in the case company. The capacity within a single simulation run is thereby assumed to be constant, as we wish to understand the interaction between the queue discipline, the capacity and the process performance. The service level constraint is included as a separate factor in order to reduce the

issue of starvation in the NPS, $SRTF$ and $LRTF$ disciplines. In this case, it is assumed that a case must have left the queue within 72 hours to be compliant. The hard ceiling therefore takes effect after 60 hours, such that agents will have 12 hours to retrieve the case from the queue. When the hard ceiling takes effect, the previous order of the cases is overruled by the $FCFS$ discipline. Further details are shown in Section 3.1.5. A total of 100 replications were performed, resulting in a total of $N = 5600$ simulation runs. The source code used to run the experiments is publicly available[1].

3.1.2. Simulation procedure

The simulation was set up in such a way as to generate event stream data from a stochastic process that is similar in its nature and properties to the one that was empirically investigated in the case company in Study 1. All algorithms were calibrated using the statistical models and distributions fitted in Study 1 (see Section 2.1.1).

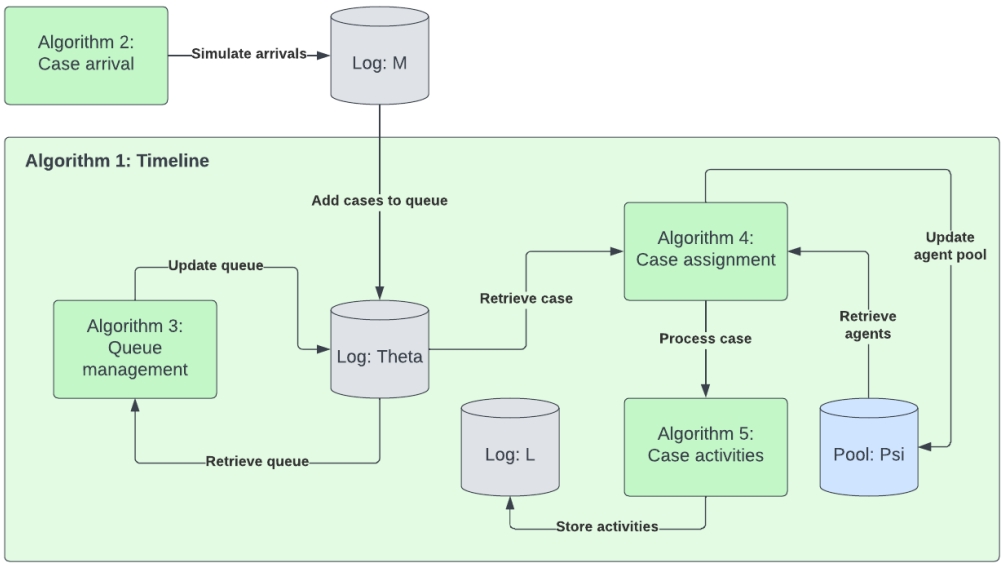

Figure 4: Overview of simulation procedure.

The general procedure of the simulation model is shown in Figure 4. The simulation model produces three event logs: M, Θ and L. Furthermore, an agent pool Ψ is defined. Firstly, the event log M is populated with all cases arriving within the simulation period using Algorithm 3. Next, the timeline is simulated using Algorithm 1. As the simulated timeline passes the dates of arrival of new cases, these are added to the event log Θ containing the cases in the queue. The queue is then updated by Algorithm 2, re-prioritising the order based on the given queue discipline. Cases in the queue are then assigned to an agent using Algorithm 4, which continuously updates the agent pool Ψ. Once a case is assigned to an agent, the case is processed using Algorithm 5 until

[1] github.com/Mikeriess/P3_queue_prioritization

the final activity is reached. The observed process behavior is then recorded in the final event log L. This process repeats itself until the end of the specified simulation period.

3.1.3. Timeline

The simulation progresses in discrete time slices (days and 15-minute intervals). Within each day and case, activities can occur if and only if three criteria are met: (a) at least one open case must have activities pending, (b) at least one agents is available for performing activities, and (c) the current day is a workday. The timeline consist of a main loop that iterates over each day within a given range D_{end}. The arrival of new cases is controlled by the inter-arrival model fitted in Section 2.2.3. When cases are closed, they will get a simulated NPS-score based on their individual progress (using Eq. 8).

Algorithm 1 is the main simulation loop that iterates over the specified simulation periods $d = (1, 2, \ldots, D)$. Firstly, the set of all cases to arrive M in the simulation period is generated using Algorithm 3. Afterwards, the status of the cases and agents is updated every 15 minutes in the following order: First, the new cases that have arrived until the present point in time z, are added to the case buffer Θ. Next, queue management as described in Algorithm 2 is performed on Θ, and cases are prioritised based on the queue discipline. Next, idle agents in the agent pool Ψ are assigned to cases that are waiting in the queue (if any) by Algorithm 4. Finally, open cases with an agent assigned are processed by Algorithm 5. The resulting activities are written to the output event log L.

3.1.4. Case arrival

A case is represented as a trace in this simulation framework. A single trace η_i is defined as a multiset $\eta = (i, q, s, t, a, j, r, c)$, consisting of attributes of varying length: the case identifier i, the arrival-time q, the status of the case s, the set of timestamps t for the performed activities, the set of activities performed in chronological order a, the set of time indexes j for the performed activities, the set of agents r working on each of the activities, and finally the static case attributes (such as *case topic*, predicted *throughput time* and *NPS*) in the set c. The arrival time of a new case is governed by the model in Equation 6, with parameter values as shown in Table 4. Implementation details are shown in pseudo-code in Algorithm 1.

3.1.5. Queue management

Depending on the prioritisation approach, the queue will be treated differently. The $FCFS$ approach will leave the current order in the queue (Θ) unchanged, whereas $SRTF$ and $LRTF$ will order the queue by their predicted throughput time at arrival in ascending and descending order, respectively. The NPS approach will re-order the cases in the queue based on their predicted NPS score, as described in Section 1.1.2. *Passives* (customers answering 7-8 in the NPS survey question) will in this case be prioritised, as passives are *as yet undecided* and therefore more likely to change, i.e. to be either positively surprised (and thereby converted to promoters) or disappointed (and thereby become detractors), given a particular amount of resource expenditure. To ensure a certain degree of fairness, we also evaluate the effect of a hard ceiling, ensuring that customers put in the back of the queue will be prioritised once they have been there for more than 60 hours. This works as a two-step approach where the sorting based on the original queue discipline is firstly performed, after this step, all customers that have waited more than 60 hours will be moved in front of the queue (increasing the likelihood of a response within an assumed service level of 72 hours). The details of the prioritisation algorithms are shown in pseudo code in Algorithm 2.

Algorithm 1: The discrete timeline simulation.

Data: queue discipline $P_{discipline} \subset \{FCFS, SRTF, LRTF, NPS\}$, number of days to simulate $D_{end} \in \mathbb{N}$, probability distributions $\omega \in \Omega$, feature mapping functions for conditional prediction models $\pi \in \Pi$, pool of case-handlers (agent objects) Ψ

Result: event log L, case-buffer Θ

```
/* Event log placeholders                                          */
```
$\Theta, M, L \leftarrow \emptyset$

```
/* Simulate the cases to arrive during the day                     */
```
$M \leftarrow CaseArrival(i, D_{end}, \pi_{arr}, \pi_{attr})$

```
/* For each day D = (1, 2, .., D_end)                              */
```
for d *in* D **do**

```
    /* Perform queue updates on a 15-minute window basis           */
```
    $z = d$

    for *every 15-minute window in day* d **do**

```
        /* Add new cases to the case pool Θ and filter out cases that
           arrived after the 15-minute window (and are therefore
           unobserved)                                             */
```
        $\Theta \leftarrow \Theta \bigcup M$

        $\Theta \leftarrow \{\eta \in \Theta, |\eta_{(q)} \leq z + 15\}$

```
        /* Sort case order, based on the queue discipline          */
```
        $\Theta_{ordered} \leftarrow QueueManagement(\Theta, P_{discipline})$

```
        /* Assign idle agents to newly arrived cases               */
```
        $\Theta, \Psi \leftarrow CaseAssignment(\Theta_{ordered}, \Psi)$

```
        /* Perform activities on active cases                      */
```
        $L, \Theta, \Psi \leftarrow CaseActivities(d, \Theta, \Psi, z)$

```
        /* Update time by 15 minutes (0.010416 days)               */
```
        $z = z + 0.010416$

```
        /* Simulate conditional NPS score for closed cases from
           Equation 8                                              */
```
        for $\eta_{(s=closed)}$ *in* Θ **do**

            $\eta_{(c=NPS_{resp})} \leftarrow Model(\eta)$

        end

    end

end

Algorithm 2: Queue management.

Data: case buffer Θ, queue discipline $P_{discipline} \subset \{FCFS, SRTF, LRTF, NPS\}$, and service level in days $P_{SLA} \in \mathbb{R}$

Result: updated case buffer $\Theta_{ordered}$

```
/* Only cases that arrived up until time z + 0.010416 are visible */
```

if $P_{discipline} = FCFS$ **then**

```
    /* Sort cases by their arrival time q */
```

$\Theta_{ordered} \leftarrow Sort(\Theta, \Theta_{(c=q)}, ascending)$

if $P_{discipline} = SRTF$ **then**

```
    /* Sort cases in ascending order by their predicted throughput
       time ŷ */
```

$\Theta_{ordered} \leftarrow Sort(\Theta, \Theta_{(c=\hat{y})}, ascending)$

if $P_{discipline} = LRTF$ **then**

```
    /* Sort cases in descending order by their predicted throughput
       time ŷ */
```

$\Theta_{ordered} \leftarrow Sort(\Theta, \Theta_{(c=\hat{y})}, descending)$

if $P_{discipline} = NPS$ **then**

```
    /* Proceed if there is more than one case and a difference in
       predicted NPS across the cases */
```

if $|\Theta| > 1$ *and* $Var(Abs(\Theta_{(c=\hat{NPS}_{arrival})} - 7.5)) > 0$ **then**

```
        /* Sort cases by their predicted NPS-score */
```

$\Theta_{ordered} \leftarrow Sort(\Theta, Abs(\Theta_{(c=\hat{NPS}_{arrival})} - 7.5), ascending)$

if $P_{SLA} > 0$ **then**

```
    /* Prioritise the sorted queue Θ_ordered further by the current
       waiting time in the queue w */
    /* Get the subset of customers where the waiting time is above
       the service level minus one day */
```

$\Theta_{priority} \subset \Theta_{(w)} \geq (P_{SLA} - 1)$

```
    /* Sort by waiting time */
```

$\Theta_{priority} \leftarrow Sort(\Theta, \Theta_{(w)}, descending)$

```
    /* Get the remaining customers */
```

$\Theta_{remaining} \subset \Theta_{(w)} < (P_{SLA} - 1)$

```
    /* Concatenate to new queue Θ_ordered, with prioritized customers
       Θ_priority, now first in the line, regardless of P_discipline */
```

$\Theta_{ordered} \leftarrow \Theta_{priority} \cap \Theta_{remaining}$

3.1.6. Case assignment

Case assignment is performed for the agent pool $\Psi = (\psi_1, \ldots, \psi_k)$, which consist of all agents in the simulated customer service unit. First, an idle-pool A is created, such that only idle agents are assigned to new cases. Then, each agent in A is assigned to cases in $\Theta_{ordered}$ that are not currently assigned to an agent. The detailed implementation can be seen in pseudo-code in Algorithm 4.

3.1.7. Case activities

Case activities are updated every 15 minutes as illustrated in Algorithm 1. In each update, each agent in the agent pool Ψ processes new activities if and only if two conditions are met: (a) the agent is assigned to a case and (b) the finish time for the current activity is within the current 15-minute window. When these conditions are met, a start delay v (Algorithm 6) is calculated if work on the next activity would be scheduled to begin outside business hours. In this case, the time until the next possible activity start *during* business hours is accumulated in v and subsequently added to the simulated activity duration. Next, Algorithm 7 simulates the next activity $a_{(j+1)}$ with timestamp $t_{(j+1)}$. The new activity and its timestamp are appended to the event log L. If the next activity is the absorbing state (END) of the Markov chain described in Section 2.2.2, the agent ψ is unassigned from the case, and the time since the agent became idle is updated, while the trace η is removed from Θ. The details of the implementation can be seen in the pseudo-code of Algorithm 5.

3.1.8. Start delay

Since an activity cannot commence when an agent is off duty, we introduce a start delay, i.e. the time between when an activity was scheduled to begin and when it could actually begin. In the simulation experiments, this start delay represents the time until business hours begin again. The business hours are defined here as Monday to Friday from 08:00 to 18:00. The details of how the start delay is generated is shown in pseudo code in Algorithm 6. We will not introduce an end delay when activities end outside business hours, as the activity duration model from Study 1 already includes these.

3.1.9. Next activity

The next activity algorithm is a subroutine that prepares the input variables needed for simulating the next activity and the duration of the next activity. After this step, the simulated values are appended to the current trace η. The next activity is generated from the Markov transition matrix described in Section 2.2.2. The activity duration is generated from the simulation model defined in Equation (3). The detailed implementation can be seen from the pseudo-code in Algorithm 7.

3.1.10. Evaluation

We answer Research Question 2.1 by measuring the average waiting time in the queue within each day d of each simulation run l. This is achieved by generating a time-series measuring the average waiting time of all customers currently in the queue at the beginning of day d, illustrated in Equation 10:

$$AvgWaitingTime = \frac{1}{N}\sum_{l=1}^{N}\frac{1}{D}\sum_{d=1}^{D}(d - H_{(q,l,d)}) \tag{10}$$

Let $H_{(q,l,d)}$ represent a vector of the individual arrival times q of all customers waiting in the queue in simulation run l at day d, as defined in Equation 11, and s denote the status of a case:

$$H \subset \{\eta \in \Theta, |\eta_{(s=waiting)}\} \tag{11}$$

In a similar manner, we answer Research Question 2.2 by calculating the average queue length, the average capacity utilisation, and the average proportion of cases closed. The average queue length is calculated using the definition in Equation 12:

$$AvgQueueLength = \frac{1}{N}\sum_{l=1}^{N}\frac{1}{D}\sum_{d=1}^{D}|H_{(i,l,d)}| \tag{12}$$

Where $H_{(i,l,d)}$ represents a list of case identifiers i for each simulation run l at day d. The average capacity utilisation is defined by Equation 13:

$$AvgCapUtilisation = \frac{1}{N}\sum_{l=1}^{N}\frac{1}{D}\sum_{d=1}^{D}\frac{|\Psi_{(j\neq\emptyset,i,d)}|}{|\Psi_{(i,d)}|} \tag{13}$$

Where $\Psi_{(d)}$ represents the full agent pool at day d, and $\Psi_{(i\neq\emptyset,d)}$ the set of agents assigned to case i on day d. Finally, the average proportion of cases closed during each individual simulation run is calculated as:

$$AvgPercentCasesClosed = \frac{1}{N}\sum_{l=1}^{N}\frac{|\Theta_{(s=closed,l)}|}{|\Theta_{(l)}|} \tag{14}$$

To answer Research Question 2.3, we first define the simulated NPS response from Equation 8 as NPS_{resp}, we then restrict the response to the discrete interval of the theoretical net promoter score [0, ..., 10] using the definition in Equation 15, and finally we round the response, resulting in $NPS_{resp} \in [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10]$ (Eq. 16).

$$NPS_{resp} = \begin{cases} 0, & x \leq 0 \\ x, & 0 > x < 10 \\ 10, & x \geq 10 \end{cases} \tag{15}$$

$$NPS_{resp} = \lceil NPS_{resp} \rceil \tag{16}$$

To calculate the average company or process-level net promoter scores NPS across all simulation runs, we use the approach proposed by [26], where $\%detractors$ denotes the proportion of customers with $NPS_{resp} < 7$ and $\%promoters$ denotes the proportion of customers with $NPS_{resp} > 8$ in the respective simulation run l:

$$AvgNPS = \frac{1}{N}\sum_{l=1}^{N}\%promoters_{(l)} - \%detractors_{(l)} \tag{17}$$

3.2. Results

3.2.1. Convergence behaviour

To assess the convergence of the simulated queuing systems to a steady state, we calculated the average queue length at the beginning (00:00) of each day in the simulation period. Figure 5 shows the convergence behaviour for all conditions without SLA, Figure 6 for all conditions with SLA. In both sets of conditions, the stabilisation of the queuing systems depended crucially on the the number of agents. While we simulated conditions ranged from three to nine agents, only those with at least seven agents reached a steady state st all.

Note that the conditional arrival rate (see Eq. 6) peaked at approximately 180 days. Since the simulated period started at 01/07/2018, this point in time corresponds to the Christmas holidays, where the conditional arrival rate had been highest in the calibration data.

3.2.2. Performance

In a next step, we compared the performance of the four queue disciplines in terms of average queue length, percent capacity utilisation, percent cases closed, average waiting time in the queue, and average case resolution time. We used linear models that included fixed main effects of queue discipline (factor levels: NPS, LRTF, SRTF, FCFS), number of agents (factor levels: 3, 4, 5, 6, 7, 8, 9) and service level agreement (factor levels: none, SLA = 60 hours), plus all two-way interactions and the three-way interaction between the factors. All models were estimated by ordinary least squares.

The number of agents in the scenario had highly significant main effects on average queue length ($F_{6,5544} = 20227.55, p < .001$), percent capacity utilisation ($F_{6,5544} = 7890.69, p < .001$), percent cases closed ($F_{6,5544} = 52078.05, p < .001$), average waiting time in the queue ($F_{6,5544} = 12372.43, p < .001$) and average case resolution time ($F_{6,5544} = 19748.65, p < .001$). Besides the number of agents, there were no other significant effects on average queue length, capacity utilisation, and percent cases closed. This indicates that case throughput and capacity utilisation depended exclusively on the number of agents available for the handling of the incoming cases; neither the queue discipline nor the presence or absence of a service level agreement had additional influences. The means are plotted in Figure 7.

The results were quite different where the duration variables were concerned. Average waiting time in the queue depended on all main effects and interactions in the experimental design, including queue discipline ($F_{3,5544} = 273.80, p < .001$), number of agents ($F_{6,5544} = 12372.43, p < .001$), SLA ($F_{1,5544} = 2442.10, p < .001$), the two-way interaction between queue discipline and number of agents ($F_{18,5544} = 96.03, p < .001$), the two-way interaction between queue discipline and SLA ($F_{3,5544} = 278.80, p < .001$), the two-way interaction between number of agents and SLA ($F_{6,5544} = 828.21, p < .001$), and the three-way interaction between queue discipline, number of agents and SLA ($F_{18,5544} = 97.06, p < .001$).

In a similar manner, average case resolution time depended on all main effects and interactions in the experimental design, including queue discipline ($F_{3,5544} = 3908.00, p < .001$), number of agents ($F_{6,5544} = 19748.65, p < .001$), SLA ($F_{1,5544} = 34552.93, p < .001$), the two-way interaction between queue discipline and number of agents ($F_{18,5544} = 621.18, p < .001$), the two-way interaction between queue discipline and SLA ($F_{3,5544} = 3854.53, p < .001$), the two-way interaction between number of agents and SLA ($F_{6,5544} = 5458.87, p < .001$), and the three-way interaction between queue discipline, number of agents and SLA ($F_{18,5544} = 596.55, p < .001$).

The means are plotted in Figure 8. Compared to FCFS, both NPS and LRTF substantially decreased case resolution time, but did so at the expense of increased waiting time. The performance of NPS and LRTF was identical. SRTF had effects in the same directions but to a lesser degree. All effects disappeared when a service level agreement was present (overriding the respective queue discipline with an FCFS regime) or when enough agents were available so that all arriving cases could be handled without delay.

3.2.3. Effects on customer NPS

In the previous two subsections, we compared the four queuing disciplines in terms of standard convergence and performance metrics. The NPS-based queue discipline developed in this paper showed the same level of performance as its closest "relative" among the competing disciplines, the longest remaining time first (LRTF) discipline. However, the rationale of the NPS-based discipline is to prioritise the cases in the queue in such a way that customer loyalty after case closure is maximised.

In terms of the average simulated individual NPS responses after case closure, all main effects and interactions in the experimental design were significant: queue discipline ($F_{3,5544} = 27.07, p < .001$), number of agents ($F_{6,5544} = 204.35, p < .001$), SLA ($F_{1,5544} = 243.74, p < .001$), the two-way interaction between queue discipline and number of agents ($F_{18,5544} = 4.02, p < .001$), the two-way interaction between queue discipline and SLA ($F_{3,5544} = 22.09, p < .001$), the two-way interaction between number of agents and SLA ($F_{6,5544} = 37.99, p < .001$), and the three-way interaction between queue discipline, number of agents and SLA ($F_{18,5544} = 2.74, p < .001$). Note that this metric has a minimum of 0 and a maximum of 10; the means are plotted in the left panel of Figure 9.

In terms of the simulated (organisation-level) NPS calculated from the distribution of the individual NPS responses, all main effects and interactions in the experimental design were significant as well: queue discipline ($F_{3,5544} = 20.73, p < .001$), number of agents ($F_{6,5544} = 147.31, p < .001$), SLA ($F_{1,5544} = 187.12, p < .001$), the two-way interaction between queue discipline and number of agents ($F_{18,5544} = 3.04, p < .001$), the two-way interaction between queue discipline and SLA ($F_{3,5544} = 15.78, p < .001$), the two-way interaction between number of agents and SLA ($F_{6,5544} = 31.46, p < .001$), and the three-way interaction between queue discipline, number of agents and SLA ($F_{18,5544} = 3.97, p < .001$). Note that this metric has a minimum of -100 and a maximum of +100; the means are plotted in the right panel of Figure 9.

Compared to FCFS, not only the NPS discipline but also LRTF and SRTF improved both target metrics. The NPS discipline, in turn, was slightly superior to LRTF and SRTF. Again, all effects disappeared when a service level agreement was present (overriding the respective queue discipline with an FCFS regime) or when enough agents were available to handle all arriving cases without delay.

3.2.4. Robustness

The above analyses were performed on the complete period of 365 simulated days. Since all simulation runs had started with an empty queue, there was higher capacity in terms of available service agents during the initial "burn-in" period before the case queue had filled up and the system had reached its steady state. To assess whether the performance differences between the four queue disciplines were robust to the presence or absence of such a burn-in period, we excluded the first 30 simulated days from all simulation runs and repeated all analyses reported above on the resulting 335-day period.

The results of all statistical tests remained stable, with the exception of the percentage of cases closed. Whilst only the main effect of number of agents had been significant in the analysis of the full 365-day period, all main effects and interactions in the experimental design were significant when we restricted the analysis to the last 335 days: queue discipline ($F_{3,5544} = 12.12, p < .001$), number of agents ($F_{6,5544} = 51652.67, p < .001$), SLA ($F_{1,5544} = 114.52, p < .001$), the two-way interaction between queue discipline and number of agents ($F_{18,5544} = 3.42, p < .001$), the two-way interaction between queue discipline and SLA ($F_{3,5544} = 16.99, p < .001$), the two-way interaction between number of agents and SLA ($F_{6,5544} = 16.88, p < .001$), and the three-way interaction between queue discipline, number of agents and SLA ($F_{18,5544} = 2.88, p < .001$). The means are plotted in Figure 10.

4. General discussion and conclusion

The aim of the research presented here was to introduce a customer service management approach with a new type of priority queuing based on predicted customer loyalty scores. In our approach, customer loyalty scores are predicted based on predicted case throughput time. Case throughput time is, in turn, predicted from seasonality indicators alone, as case attributes are not available at the time of entering the queue.

We presented two studies to calibrate and test our approach. Study 1 was a statistical analysis of two years of empirical customer service data from a European internet and telecommunications provider. We utilised these real-world data to obtain ecologically valid estimates of the parameters of all distributions governing the customer service process. In Study 2, we used the calibrated distribution parameters in Monte Carlo simulations of the customer service management system, comparing our predictive priority queuing approach with the traditional first come, first served approach (FCFS) and two alternative prioritisation approaches based on predicted throughput time; the longest remaining time first approach (LRTF) and the shortest remaining time first approach (SRTF).

4.1. Key findings

The results indicate that prioritisation based on predicted increases in customer loyalty does indeed lead to an increase in average customer loyalty relative to FCFS, SRTF and LRTF. In addition, it also leads to lower average case throughput time, but only compared to FCSF and SRTF; LRTF performed equally well in this respect as the NPS approach proposed here. Both effects are strong when the number of customer service agents is low but disappear when the number of customer service agents is so high that all cases can be processed immediately. In our opinion, this is the key finding from our Monte Carlo studies. The novel approach we suggested in this paper seems to have most leverage in a service organisation with little excess capacity. The ”leaner” the organisation, the more it stands to gain from introducing the prioritisation approach we suggest.

Interestingly, the other two queue disciplines with case prioritisation based on predicted throughput time (i.e., LRTF and SRTF) showed very similar performance patterns as our NPS-based approach. Indeed, the performance of the LRTF discipline was so similar to that of the NPS-based discipline that their difference was not statistically significant on most performance indicators.

Although striking when regarded superficially, the similarity of the behaviour of LRTF and NPS is not really surprising when one considers how cases are prioritised under the two disciplines. Both share an identical first prediction step (predicted throughput time; see Eq. 7). The

LRTF discipline prioritises the cases directly based on this prediction. NPS takes the prediction as input to another prediction (NPS; see Eq. 9). However, this second model is linear and does not affect the priority order of the cases. Only the last step in the calculation of the priority rank is non-linear (see Eq. 1), assigning priority ranks to the cases as a function of their proximity to the mid-point of the region defining the ”passives” on the scale of individual NPS responses. At least in theory - in practice, however, a large majority of the cases ended up on the same side of that mid-point (because a large majority of the customers in the calibration data had been ”promoters” and the distribution was assumed to be the same in the simulation experiments), rendering also the last transformation step predominantly linear and therefore leaving the priority order largely unaffected. In follow-up studies, the distribution of the individual NPS responses should be systematically varied (as opposed to be kept constant and equal to the distribution observed in the calibration data, as was done in the present research) to identify the conditions under which the NPS-based prioritisation discipline will diverge from the LRTF discipline.

Another result that appears striking at first but is not all that surprising when considered more closely, is the disappearance of all throughput time differences between the four queue disciplines when a service level of 60 hours was introduced in order to avoid starvation affects and thereby make the priority-based queue disciplines fairer (see Fig. 8). The way our queuing systems tried to fulfil this constraint was by expediting cases that had been waiting for more than 60 hours to the front of the queue, and if there were several such cases, in the order in which they had entered the queue. Since this way of re-ordering the queue is equivalent to the FCFS discipline, it ”overrides” the priority-based disciplines with FCFS. Hence, all performance differences disappeared, at least with the rather stringent service level of 60 hours we applied in Study 2, which had been motivated by the service level agreements currently in use at the case company who had supplied us with the calibration data. In order to reap the benefits of the priority-based queue disciplines investigated here but still live up to elementary levels of fairness, somewhat less stringent service levels should be set. In future research, the service level should be systematically varied in order to identify a level that can be regarded as a reasonable compromise.

4.2. Limitations

In addition to the limitations already discussed above, there were some issues related to the calibration data which we would like to discuss here.

One aspect of Study 1 that might be criticised is the possibility that a systematic selection effect led to the composition of the case and customer sample which we used to train our models and calibrate our simulations. As we described in detail in Section 2.1.1, we only used data from service cases where the customers had actually responded to an SMS request to answer the NPS question. One could argue in the tradition of the exit, voice and loyalty model [37, 38, 39] that customers who respond to such requests are on the ”voice” side, suggesting that they have a stronger commitment to the relationship with their service provider. However, we do not consider it plausible that the effects in our models would have opposite signs if they were estimated among customers with weaker average commitment. If the effects were weaker but the signs remained the same, our models would still produce similar (albeit noisier) results.

A second limitation is that the calibration data used in Study 1 did not include cases created via phone, as these are not entered into the queue but processed immediately. The prediction models fitted in Study 1 and the simulations reported in Study 2 did therefor not represent this type of case. However, the case company informed us that such cases constitute only a minor proportion of all cases. In addition, we have no information from the case company or any other

reason to assume that such cases occur in a systematic manner at specific times such that they would temporarily overload the queue. Hence, we consider our results to be robust against this issue.

A third limitation related to the calibration data in Study 1 is that the case company did not provide information about case abandonment or re-assignment. In the simulations, we therefore assumed the queue prioritisation to be non-preemptive at all times. But even if this assumption was violated, this would merely mean that the mean and the variance of case completion time observed in the calibration data was due to service agents temporarily switching between cases. In Study 2, the bias would have affected the simulations of the four competing queue disciplines in the same way and should therefore not have has an influence on the performance differences *between* the disciplines.

Finally, since the calibration data were obtained from a specific case company in a specific observation period, Study 1 shares the limitations of all case studies: the variation of all features of the service process was limited to the range of variation observed in the calibration data. Hence, the simulations in Study 2 had the great advantage that their parameters **were** based on real-life calibration data, as opposed to assumptions alone. On the other hand, this also had the consequence that the range of scenarios we considered in Study 2 was limited to what appeared reasonable to us in the context of the case company and other companies similar to it. Follow-up studies with systematic variation to factors that has been kept constant here (including variation to more extreme values than used in this case study), will shed additional light on the behaviour of the investigated queue disciplines, and thereby increase the generalisability of our research.

4.3. Applicability

Strictly speaking, the applicability of our approach to service processes in other organisations depends on one condition: customer loyalty must be inversely related to waiting time. This relationship has to be monotonous, but it does not necessarily have to be strong. In the empirical data we used for calibration (Study 1), the squared multiple correlation between case throughput time and NPS was no more than $R^2 = 0.04$. This is clearly sufficient to obtain strong performance improvements relative to a traditional first come, first served approach (as shown in Study 2). Other empirical studies of the relationship between waiting time and customer satisfaction or loyalty in different types of service settings obtained similar effect sizes [28, 29, 1, 31]. Hence, our approach should normally be applicable in a similar form in other service settings.

Although our approach may in principle be applicable in most customer service settings, its feasibility depends on the availability of case features (e.g., extracted from the CRM system of the organisation) that allow an early prediction of case throughput or waiting time. In the empirical data we used for calibration of our models, no content-related case features were used. The models we used to predict case throughput time (and based on that, NPS gain) only utilised seasonality features - that is, information about the inter-temporal variation of case load and server capacity - but still performed sufficiently well. If this can be generalised to other service settings, and possibly even improved by including content-related case features if these are available, feasibility should not be a major problem.

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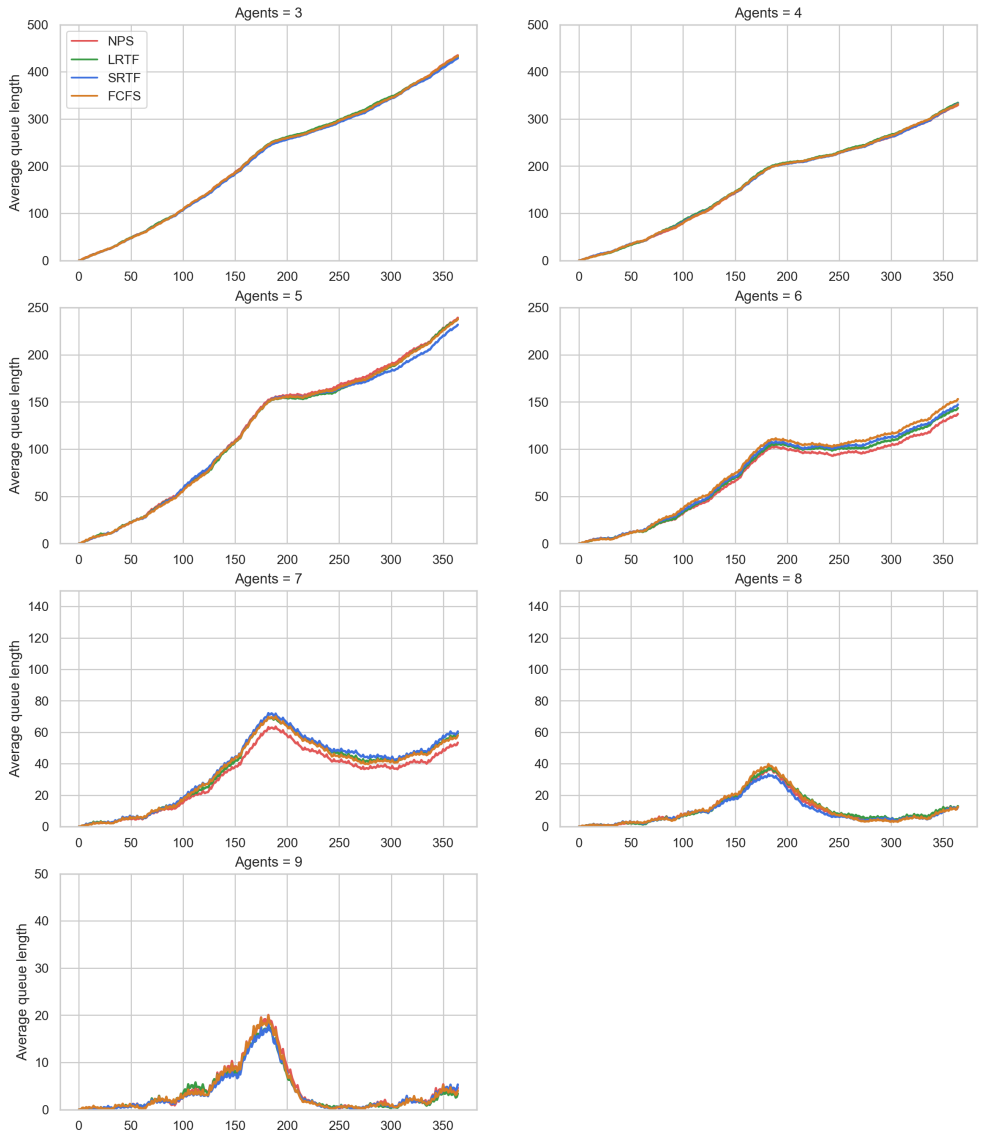

Figure 5: Average daily queue length as a function of simulation day and number of agents (no SLA). **Note:** the range of the y-axis is different for each row of diagrams.

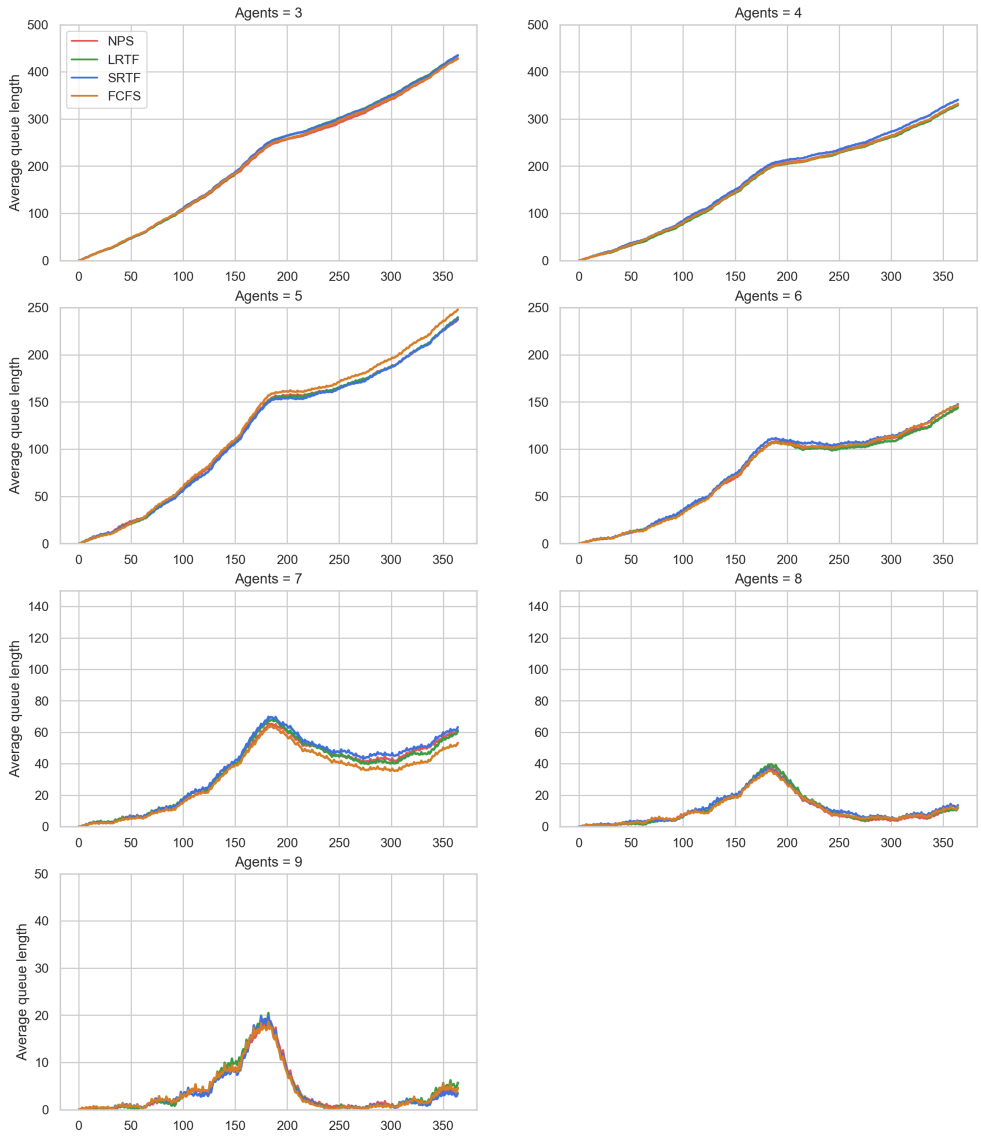

Figure 6: Average daily queue length as a function of simulation day and number of agents (SLA = 60 hours). **Note:** the range of the y-axis is different for each row of diagrams.

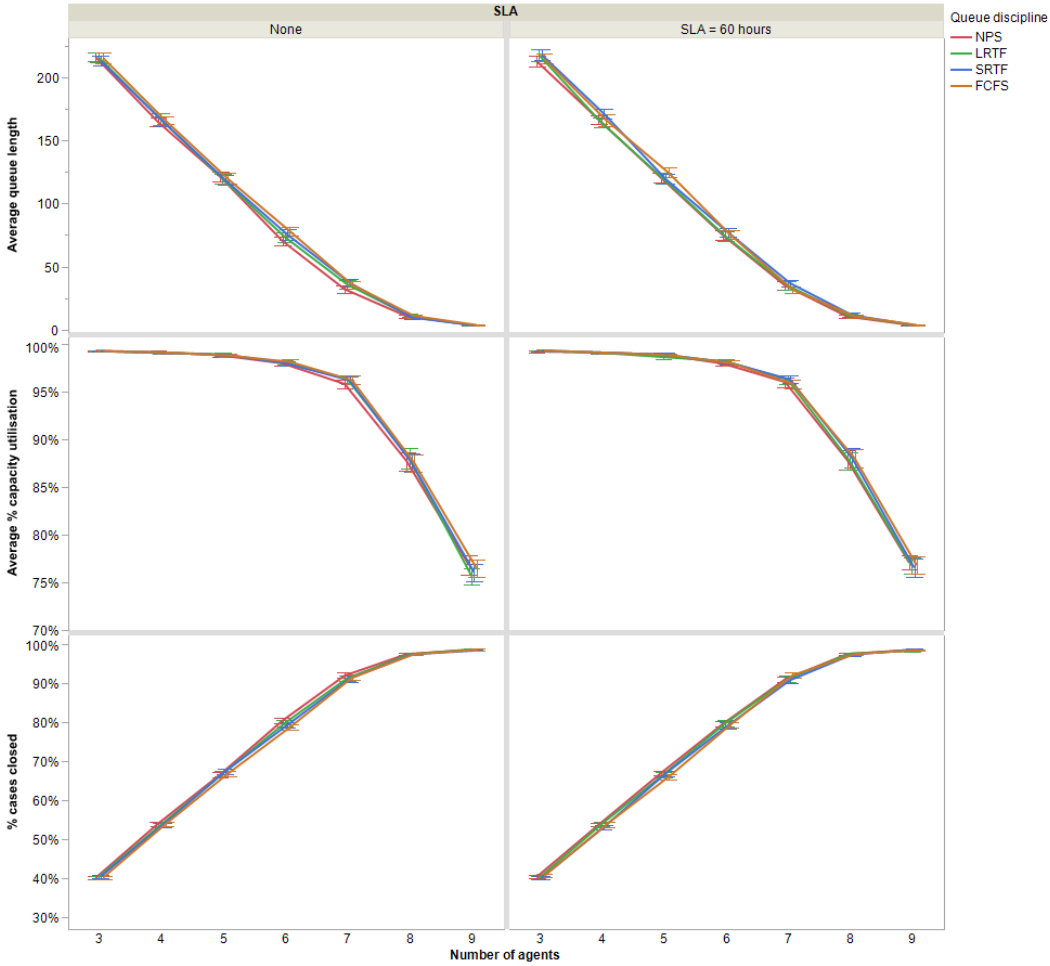

Figure 7: Average queue length, percent capacity utilisation, and percent cases closed as a function of queue discipline, number of agents, and service level agreement (error bars represent 95% confidence intervals).

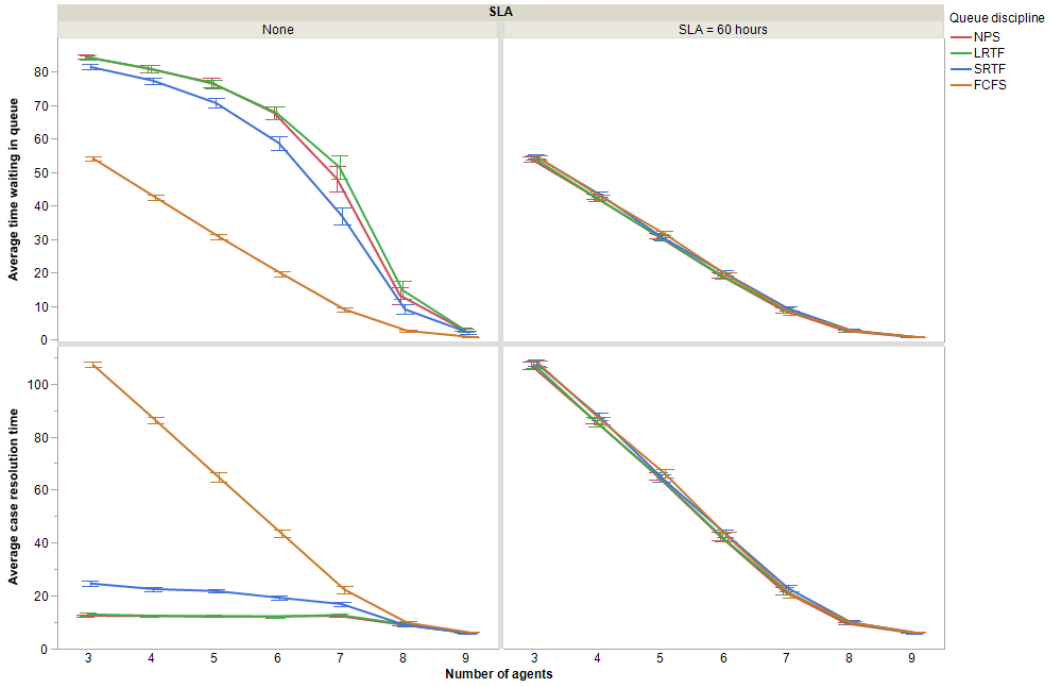

Figure 8: Average time waiting in queue and average case resolution time as a function of queue discipline, number of agents, and service level agreement (error bars represent 95% confidence intervals).

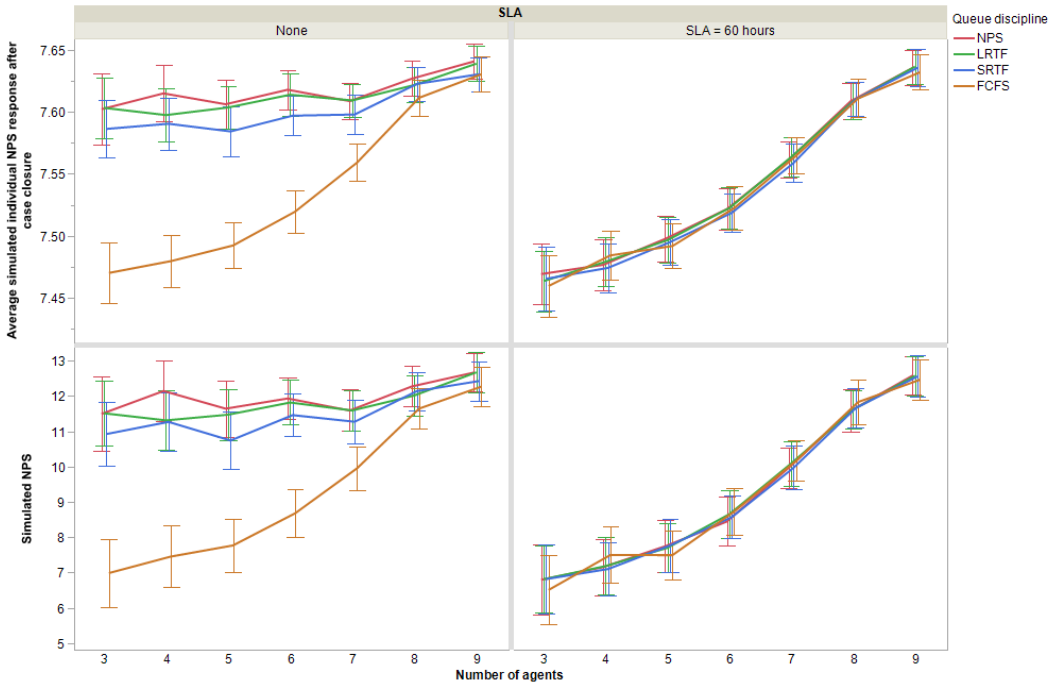

Figure 9: Average simulated individual NPS response after case closure and simulated NPS (on organisation level) as a function of queue discipline, number of agents and service-level agreement (error bars represent 95% confidence intervals).

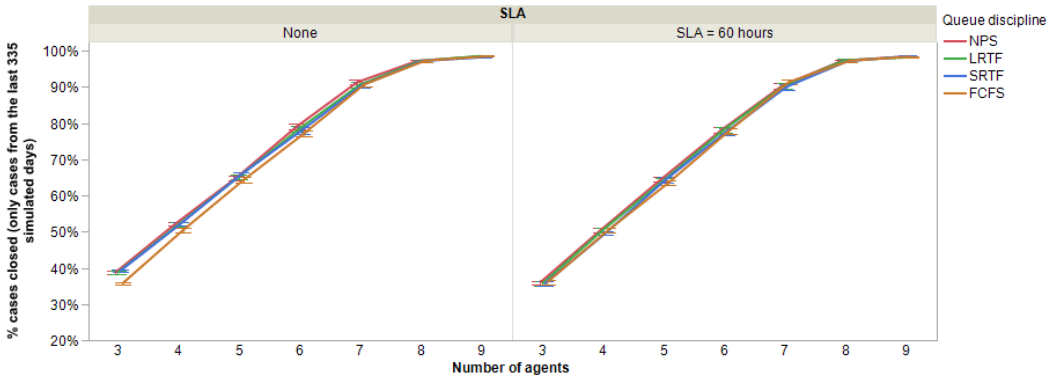

Figure 10: Percent cases closed during the last 335 simulated days as a function of queue discipline, number of agents and service-level agreement (error bars represent 95% confidence intervals).

Appendix A. Algorithm: Case arrival

Algorithm 3: Case arrival.

Data: case identifier starting value i, number of days to simulate D_{end}, feature mapping function for inter-arrival time model π_{arr}, probability distribution for case attributes ω_{attr}

Result: set of newly arrived cases M, updated case identifier value i

```
/* Initialize the time variable z, and set of cases M          */
```
$z = 0$

$M \leftarrow \emptyset$

```
/* Simulate today's case arrivals                               */
```
while *current time* $z < D_{end} + 1$ **do**

```
    /* Simulate inter-arrival time q, from model in Equation 6  */
```
$X \leftarrow \pi_{arr}(z, d)$

$q = -\log(1 - U(0,1)) \times \exp(\alpha + \sum_{i=1}^{n} X_i \beta_i)$

```
    /* Generate a new case with arrival time q and assign a case
       identifier i                                             */
```
$\eta \leftarrow (i, q, s \leftarrow \emptyset, t \leftarrow \emptyset, a \leftarrow \emptyset, j \leftarrow \emptyset, r \leftarrow \emptyset, c \leftarrow \emptyset)$

```
    /* Predict throughput time of the case from Equation 7      */
```
$\eta_{(c=throughput)} \leftarrow Model(\eta_{(c=\{1,\ldots,k\})})$

```
    /* Predict NPS score of the case from Equation 9            */
```
$\eta_{(c=\hat{NPS}_{arrival})} \leftarrow Model(\eta_{(c=throughput)})$

```
    /* Append the case η to the set of cases M                  */
```
$M \leftarrow M \bigcup \eta_i$

```
    /* Update time and case identifier variables                */
```
$z = z + q$

$i = i + 1$

end

Appendix B. Algorithm: Case assignment

Algorithm 4: Case assignment.

```
Data: ordered case buffer Θ_ordered, pool of agents Ψ = (ψ_1, ..., ψ_k)
Result: pool of agents Ψ
/* Create a temporary pool for idle agents A                    */
A ← ∅
/* For every agent in the agent pool                            */
for ψ in Ψ do
    /* If agent is not assigned to a case                       */
    if ψ_(i) ≠ ∅ then
        /* Add agent to idle pool, remove from agent pool       */
        A ← ψ
        Ψ ← Ψ \ ψ
        /* Sort agents in idle pool by time last active t       */
        A ← Sort(A, A_(t), ascending)
end
/* For every agent in the idle pool A                           */
for ψ in A do
    /* For every case in the ordered case buffer Θ_ordered      */
    for η in Θ_ordered do
        /* If case is currently unassigned                      */
        if η_(r=∅) then
            /* If agent is still unassigned                     */
            if ψ_(i=∅) then
                /* Update case resource with agent identifier k */
                η_(r) ← ψ_(k)
                /* Assign agent to the case via case identifier i */
                ψ_(i) ← η_(i)
    end
end
/* Append all agents to agent pool again                        */
Ψ ← Ψ ⋃ A
```

Appendix C. Algorithm: Case activities

Algorithm 5: Case activity simulation

```
Data: Case buffer Θ, pool of agents Ψ, event log L, current time window start z
Result: event log L, case buffer Θ, pool of agents Ψ
/* For each agent                                                    */
for ψ in Ψ do
    /* Assigned to a case                                            */
    if ψ_(i) ≠ ∅ then
        /* Get the active case of the agent                          */
        η ← Θ_(η|i=ψ_(i))
        /* Get finish-time of current activity, and next time step k
           */
        y ← max(η_(t)), k ← max(η_(j)) + 1
        /* While y is still within the current 15-minute window      */
        while y < z + 0.010416 do
            /* Execute Alg. 6 and add a conditional start-delay      */
            v = StartDelay(y)
            /* Execute Alg. 7 to generate next activity and timestamp
               */
            η, y ← NextActivity(η, k, v, φ_act, φ_dur, π_dur)
            /* If absorption state is reached: unassign agent, update
               last activity time and remove case from Θ             */
            if a_(j+1) = END then
                ψ_(i) ← ∅
                ψ_(t) ← max(η_(t))
                Θ ← Θ \ η
            /* Append activity j (and its attributes) to event log L,
               and its appropriate trace in case buffer Θ            */
            L ← L ⋃ η_(j=k)
            Θ_(i=η_(i)) ← η_(j=k)
            /* Update agent pool Ψ with new information for agent ψ */
            Ψ ← Update(Ψ, ψ)
        end
end
```

Appendix D. Algorithm: Start delay

Algorithm 6: Start delay subroutine

```
Data: Finish time for last activity y
Result: Delay in days v
/* Define the total delay                                          */
v = 0
/* Get weekday:  {1, 2, ..., 7} ∈ ℕ                                */
weekday ← GetDayOfWeek(y)
/* Get time of day:  {1, 2, ..., 24} ∈ ℕ                           */
timeofday ← GetTimeOfDay(y)
if weekday > 5 then
    /* Add weekday delay:  days until Monday plus time until 8:00  */
    v = (8 − (weekday + 24/timeofday)) + 0.33334
else if then
    /* If tomorrow is not Saturday:                                */
    if weekday < 5 then
        /* If within work week but before business hours, add delay
           until 8:00                                              */
        if timeofday < 8 then
            v = 0.33334 − 24/timeofday
        /* If within work week but after business hours, add delay
           until tomorrow                                          */
        if timeofday > 18 then
            v = 1 − 24/timeofday + 0.33334
    /* If tomorrow is Saturday:                                    */
    else if then
        /* If within work week but before business hours, add delay
           until 8:00                                              */
        if timeofday < 8 then
            v = 0.33334 − 24/timeofday
        if timeofday > 18 then
            v = (8 − (weekday + 24/timeofday)) + 0.33334
```

Appendix E. Algorithm: Next activity

Algorithm 7: Next activity subroutine

Data: trace η, next time step k, activity start delay v, conditional activity-sequence model ϕ_{act}, conditional activity duration model ϕ_{dur}, feature mapping function for duration model π_{dur}

Result: Updated trace η, finish time of current activity y

```
/* Get last activity and timestamp                                  */
```
$a_j \leftarrow \eta_{(a|j=k-1)}$

$t_j \leftarrow max(\eta_t)$

```
/* Simulate the next activity from Equation 4 (Markov chain)        */
```
$a_{(j+1)} \leftarrow \phi_{act}(a_{(j+1)}|a_j)$

/* Simulate duration of $a_{(j+1)}$ from Equation 3, using feature mapping π_{dur}, and add start delay v to overall duration $t_{(j+1)}$ */

$X = \pi_{dur}(\eta)$

$t_{(j+1)} = \phi_{dur}(t_{(j+1)}|X) + v$

```
/* Append new activity and timestamp to trace                       */
```
$\eta_{(a)} \leftarrow a_{(j+1)}$

$\eta_{(t)} \leftarrow t_{(j+1)}$

```
/* Update finish time for current activity                          */
```
$y \leftarrow t_{(j+1)}$

Appendix F. Activity duration: Model selection

Table F.8: Model comparison: Activity duration (ML: Maximum likelihood, OLS: Ordinary least squares)

Distribution	**Method**	**Validation**	**# Param**	**Train R^2**	**Valid R^2**
Normal	OLS	None	27	0.04475	
Normal	Lasso	Holdback	15	0.0465	0.0373
Normal	Elastic Net	Holdback	27	0.03529	0.05951
Normal	Ridge	Holdback	28	0.01891	0.0121
LogNormal	ML	None	27	0.07969	
LogNormal	Lasso	Holdback	26	0.08224	0.07251
LogNormal	Elastic Net	Holdback	23	0.08420	0.06674
LogNormal	Ridge	Holdback	28	0.07633	0.08477
Weibull	ML	None	27	0.21084	
Weibull	Lasso	Holdback	7	0.17523	0.08078
Weibull	Elastic Net	Holdback	21	0.21347	0.19989
Weibull	Ridge	Holdback	28	0.21048	0.19185

Appendix G. Activity duration: All model parameters

Table G.9: Part 1: Activity duration: All model parameters.

Term	Notes	$\hat{\beta}$	SE	p
Intercept		1.6645	0.1650	¡.0001
case_topic[d_2-z_4]		0.0200	0.0456	0.662
case_topic[g_1-z_4]		-0.0538	0.0313	0.0857
case_topic[j_1-z_4]		-0.0557	0.0213	0.0088
case_topic[q_3-z_4]		0.1712	0.0587	0.0035
case_topic[r_2-z_4]		0.0836	0.0383	0.0288
case_topic[w_1-z_4]		-0.0609	0.0286	0.0334
case_topic[w_2-z_4]		0.0119	0.0313	0.7043
case_topic[z_2-z_4]		-0.0420	0.0433	0.3319
case_topic[z_3-z_4]		0.1637	0.0369	¡.0001
activity[Email-Task-Reminder]		0.0180	0.1158	0.8767
activity[Interaction-Task-Reminder]		0.1057	0.1319	0.4229
activity_number		0.0420	0.0177	0.0176
resource[a_3-z_4]	Forced in	0.1054	0.0962	0.2731
resource[a_5-z_4]	Forced in	0.0923	0.0831	0.2667
resource[a_6-z_4]	Forced in	0.0852	0.0975	0.3821
resource[a_7-z_4]	Forced in	0.2364	0.0860	0.006
resource[a_8-z_4]	Forced in	0.0508	0.0783	0.517
resource[a_9-z_4]	Forced in	0.1610	0.0843	0.0563
resource[a_10-z_4]	Forced in	0.1152	0.0770	0.1348
resource[b_3-z_4]	Forced in	0.0774	0.0895	0.387
resource[b_4-z_4]	Forced in	-0.0882	0.1011	0.383
resource[b_5-z_4]	Forced in	0.8784	0.3453	0.011
resource[b_6-z_4]	Forced in	0.0893	0.0856	0.2967
resource[b_7-z_4]	Forced in	0.3137	0.1395	0.0245
resource[b_8-z_4]	Forced in	-0.0646	0.0819	0.4304
resource[c_2-z_4]	Forced in	0.2095	0.1011	0.0383
resource[c_3-z_4]	Forced in	0.2009	0.0955	0.0354
resource[d_1-z_4]	Forced in	0.0000	0.0000	1

Table G.10: Part 2: Activity duration: All model parameters.

Term	Notes	$\hat{\beta}$	SE	p
resource[e_4-z_4]	Forced in	0.0603	0.0805	0.454
resource[e_5-z_4]	Forced in	0.0896	0.0852	0.2929
resource[e_6-z_4]	Forced in	0.0874	0.0771	0.2566
resource[e_7-z_4]	Forced in	0.1154	0.0833	0.1658
resource[f_2-z_4]	Forced in	0.1227	0.0776	0.1141
resource[f_3-z_4]	Forced in	3.3625	0.2821	¡.0001
resource[f_4-z_4]	Forced in	0.4485	0.0820	¡.0001
resource[g_9-z_4]	Forced in	-0.0384	0.0836	0.6463
resource[h_4-z_4]	Forced in	0.2234	0.0943	0.0178
resource[h_5-z_4]	Forced in	0.2686	0.1076	0.0126
resource[i_7-z_4]	Forced in	-0.1436	0.0874	0.1002
resource[i_9-z_4]	Forced in	-0.1067	0.0805	0.1851
resource[i_11-z_4]	Forced in	0.1952	0.0913	0.0326
resource[i_12-z_4]	Forced in	0.0271	0.0846	0.7486
resource[j_3-z_4]	Forced in	0.1261	0.0851	0.1383
resource[j_4-z_4]	Forced in	0.0915	0.0793	0.2487
resource[j_5-z_4]	Forced in	0.0673	0.0833	0.4189
resource[j_6-z_4]	Forced in	0.0833	0.0806	0.3011
resource[j_7-z_4]	Forced in	0.0993	0.0940	0.291
resource[j_8-z_4]	Forced in	0.0984	0.0780	0.2071
resource[j_9-z_4]	Forced in	0.0467	0.0774	0.5465
resource[j_10-z_4]	Forced in	0.3716	0.1367	0.0065
resource[l_4-z_4]	Forced in	-0.0284	0.0786	0.7178
resource[l_5-z_4]	Forced in	0.0216	0.0820	0.7923
resource[l_6-z_4]	Forced in	0.7087	0.2148	0.001
resource[l_7-z_4]	Forced in	0.1292	0.0801	0.1068
resource[m_6-z_4]	Forced in	0.0067	0.1011	0.947
resource[m_7-z_4]	Forced in	3.0204	0.3968	¡.0001
resource[n_3-z_4]	Forced in	0.1794	0.0841	0.0329
resource[n_4-z_4]	Forced in	0.1639	0.0866	0.0584

Table G.11: Part 3: Activity duration: All model parameters.

Term	Notes	$\hat{\beta}$	SE	p
resource[n_5-z_4]	Forced in	0.0328	0.0759	0.6655
resource[p_4-z_4]	Forced in	0.1128	0.0811	0.1644
resource[p_5-z_4]	Forced in	0.1177	0.0781	0.1318
resource[p_6-z_4]	Forced in	0.0702	0.0790	0.3742
resource[p_7-z_4]	Forced in	0.2800	0.0901	0.0019
resource[q_3-z_4]	Forced in	-0.0533	0.1375	0.6986
resource[q_4-z_4]	Forced in	0.0172	0.0844	0.8384
resource[r_2-z_4]	Forced in	0.1459	0.0892	0.1018
resource[r_3-z_4]	Forced in	0.1851	0.0924	0.0452
resource[r_4-z_4]	Forced in	0.0482	0.0880	0.5838
resource[r_5-z_4]	Forced in	0.0932	0.0758	0.2188
resource[r_6-z_4]	Forced in	0.0661	0.0763	0.3857
resource[r_7-z_4]	Forced in	0.0686	0.0761	0.3672
resource[r_9-z_4]	Forced in	0.0918	0.0801	0.2517
resource[s_1-z_4]	Forced in	-0.0159	0.0883	0.8568
resource[s_2-z_4]	Forced in	0.0720	0.0808	0.3725
resource[t_2-z_4]	Forced in	0.4989	0.1339	0.0002
resource[t_3-z_4]	Forced in	0.1595	0.0835	0.0559
resource[t_4-z_4]	Forced in	0.0752	0.0816	0.3566
resource[u_4-z_4]	Forced in	0.0798	0.0853	0.3497
resource[w_6-z_4]	Forced in	0.0901	0.0924	0.3296
resource[w_7-z_4]	Forced in	0.3596	0.1477	0.0149
resource[y_3-z_4]	Forced in	0.0573	0.0922	0.5344
resource[y_4-z_4]	Forced in	0.5338	0.1891	0.0048
resource[y_5-z_4]	Forced in	0.1264	0.0812	0.1197
resource[y_6-z_4]	Forced in	0.2045	0.0903	0.0236
resource[y_7-z_4]	Forced in	0.2302	0.0991	0.0202
resource[y_8-z_4]	Forced in	0.1879	0.0977	0.0545
resource[y_9-z_4]	Forced in	0.0906	0.0869	0.2968
resource[z_3-z_4]	Forced in	0.1188	0.0828	0.1514

Appendix H. Inter-arrival time: Model selection

Table H.12: Model comparison: Arrival rate. ML: Maximum likelihood, OLS: Ordinary least squares.

Distribution	**Method**	**Validation**	**# Param**	**Train R^2**	**Valid R^2**
Exponential	ML	None	5	0.3367	
Exponential	Lasso	Holdback	4	0.3823	0.1739
Exponential	Elastic Net	Holdback	5	0.3338	0.3417
Weibull	ML	None	6	0.0668	
Weibull	Lasso	Holdback	5	0.0687	0.0520
Weibull	Elastic Net	Holdback	6	0.0654	0.0525
Normal	OLS	None	6	0.0162	
Normal	Lasso	Holdback	2	0.0000	0.0000
Normal	Elastic Net	Holdback	6	0.0129	0.0206

Appendix I. Throughput time: Model selection

Table I.13: Model comparison: Throughput time (ML: Maximum likelihood, OLS: Ordinary least squares).

Distribution	**Method**	**# Param**	**Train R^2**	**Valid R^2**
Normal	Adaptive Lasso	4	0.0047	0.0047
Normal	Adaptive Elastic Net	4	0.0047	0.0047
Normal	Lasso	4	0.0046	0.0047
Normal	Elastic Net	7	0.0048	0.0038
Exponential	Adaptive Lasso	5	0.0558	0.0532
Exponential	Adaptive Elastic Net	5	0.0558	0.0532
Exponential	Lasso	6	0.0571	0.0466
Exponential	Elastic Net	3	0.0539	0.0460
Weibull	Adaptive Lasso	6	0.0067	0.0187
Weibull	Adaptive Elastic Net	6	0.0067	0.0186
Weibull	Lasso	7	0.0069	0.0175
Weibull	Elastic Net	7	0.0069	0.0175

Appendix J. NPS: Model selection

Table J.14: Model comparison: NPS simulation model.

Distribution	**Method**	**# Param**	**Train R^2**	**Valid R^2**
Normal	Lasso	12	0.03527	0.03738
Normal	Elastic Net	12	0.03527	0.03738
Gamma	Lasso	10	0.04059	0.03969
Gamma	Elastic Net	10	0.04059	0.03969
Exponential	Lasso	10	0.00195	0.00210
Exponential	Elastic Net	10	0.00195	0.00210

Table J.15: Model comparison: NPS prediction model.

Distribution	**Method**	**# Param**	**Train R^2**	**Valid R^2**
Normal	Adaptive Elastic Net	3	0.01320	0.03208
Normal	Adaptive Lasso	3	0.01320	0.03208
Normal	Lasso	3	0.01320	0.03208
Exponential	Adaptive Elastic Net	2	0.00074	0.00182
Exponential	Lasso	2	0.00074	0.00182
Exponential	Adaptive Lasso	2	0.00074	0.00182

Appendix K. Results: NPS interaction effects

Table K.16: Least squares means: average actual customer NPS as a function of case prioritisation approach and number of available customer service agents

Factor level combination	Least squares mean	SE
FCFS, 3 agents	8.8872	0.0786
FCFS, 5 agents	8.9669	0.0256
FCFS, 10 agents	9.2428	0.0512
FCFS, 15 agents	9.2499	0.0507
NPS, 3 agents	9.0665	0.0775
NPS, 5 agents	9.1093	0.0254
NPS, 10 agents	9.2421	0.0505
NPS, 15 agents	9.2637	0.0510

Appendix L. Results: Throughput time interaction effects

Appendix M. Results: Cases closed interaction effects

Table L.17: Maximum likelihood means with logarithmic link function, assuming an exponential distribution: average actual throughput time as a function of case prioritisation approach and number of available customer service agents

Priority scheme	Number of agents	Estimate	Std Error
FCFS	3	4.6741	0.1000
FCFS	5	4.1828	0.1000
FCFS	10	1.6644	0.1000
FCFS	15	1.4705	0.1000
NPS	3	2.5700	0.1000
NPS	5	2.5488	0.1000
NPS	10	1.6626	0.1000
NPS	15	1.4643	0.1000

Table M.18: Least squares: Priority scheme vs. Number of agents. F-test P. = 0.6366

Level	Least Sq Mean	Std Error
FCFS, 3	291.05000	2.46556
FCFS, 5	483.19000	2.46556
FCFS, 10	720.26000	2.46556
FCFS, 15	715.69000	2.46556
NPS, 3	292.07000	2.46556
NPS, 5	482.37000	2.46556
NPS, 10	717.45000	2.46556
NPS, 15	719.04000	2.46556